Pranjalya Tiwari

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Profile

Highly motivated and results-driven Software Engineer in Machine Learning with over 4 years of experience in developing and deploying scalable AI solutions. Proven ability to leverage deep learning and cloud computing to solve complex business problems. Passionate about building innovative technologies and driving efficiency through automation. Seeking to contribute expertise and leadership to a challenging role in a forward-thinking organization.

Education

Bhilai Institute of Technology — Durg, CG
B.Tech. Computer Science and Engineering *GPA: 9.27* — August 2017 - June 2021

Experience

Ksolves India Ltd. — Noida, UP
Senior Machine Learning Engineer — January 2025 - Present

- For **Nokia**, built an Agentic Langgraph and Langchain based application to help Nokia Consultants and Care Program Managers generate reports and insights regarding Customers from external and internal sources.
- Modified **Langflow** for **Nokia's** internal users to build and prototype AI agents. Incorporated additional features such as Single Sign-On, Role-Based Access Control, and a productionization system for the prototype agents.
- Built and deployed for production a data analysis and question answering system for **Nokia** internal team to help save time on PowerBI reports analysis, based on **Text-to-Code** system using Langgraph and Chainlit.

Deepsync Technologies / Dubpro.ai — Noida, UP
Machine Learning Engineer — July 2021 - December 2024

- Developed and deployed production-level, scalable audio, vision, and NLP solutions, utilizing machine learning and deep learning techniques to address diverse business challenges.
- Led the development and implementation of an end-to-end dubbing pipeline using Apache Airflow, capable of processing hundreds of videos concurrently, optimizing the dubbing process for efficiency and scalability.
- Designed and developed interactive demos for various models using Huggingface's Gradio, showcasing the capabilities and potential applications of the models to stakeholders.
- Optimized company expenditures by identifying and replacing costly components with efficient, open-source alternatives, resulting in significant cost savings and improved accuracy.

Backprop — Remote
Software Engineer — Oct 2020 - June 2021

- Applied Reinforcement Learning principles to solve dynamic pricing problems, enhancing pricing strategies and improving revenue generation.
- Developed a custom script to streamline data transfer between Trello and Jira, improving project management efficiency and data organization.

Skills

Technical Skills:	Deep Learning, DevOps, MLOps, Cloud Computing, Data Analysis, Bash Scripting, Data Extraction, Software Development, Web Development, Full-Stack Development
Technologies:	Python, PyTorch, Golang, SQL, NoSQL, Docker, Tensorflow, AWS, GCP, Azure, Langchain, Vector DB, Flask, REST APIs, NVIDIA Triton JavaScript, Node.js, Vue.js, Solidity, Next.js
Soft Skills:	Leadership, Problem-solving, Communication, Time management, Adaptability, Teamwork and collaboration, Listening skills
Certifications:	Deep Learning Specialization (Coursera), Data-Driven Astronomy (Coursera), Programming in `C++` (Gold medal, NPTEL), Reinforcement Learning (University of Alberta) GCP Big Data and Machine Learning Fundamentals (Coursera)

Projects

Ghazal / Poem Retrieval System *Python, PyTorch* link
A Ghazal / Poem Retrieval system built using Next.js (frontend) and Flask (backend) with SQLite as database, powered by Meta's LLama3.2 (LLM) hosted locally using Ollama and Langchain and Qdrant (Vector DB) for semantic search and retrieval, in Docker container. It also has an optional RAG approach for generating new Sher and Ghazals based on this retrieval system, using fine-tuned embeddings.

Multi-lingual NLP Service Deployment on Triton Server *Python, Docker, Azure, MLOps*
Deployed a containerized, scalable, multi-lingual NLP service using NVIDIA Triton Inference Server. Supported NER, text classification, summarization, language detection, and key phrase extraction across major languages. Leveraged Docker and Azure for deployment.

Multi-Stage LLM Pipeline for Conversational Chatbot Response Generation *Python, LLMs*
Developed a chatbot response generation pipeline using multiple LLMs and emotion classification. Integrated emotion analysis and used LLMs like Facebook's LLAMA 7b for response refinement, enhancing chatbot engagement.

Optimized AI Image Processing Pipeline with Stable Diffusion and SAM *Python, PyTorch, AWS*
Enhanced an AI image processing pipeline using Stable Diffusion and Segment Anything (SAM). Optimized model inference time and reduced memory leaks, resulting in improved performance and efficiency.

Self-hosted Gradio Demo for TorToiSe TTS *Python, PyTorch* link
Created a Gradio demo with a customized UI for local self-hosting of TorToiSe TTS, enabling voice creation and diverse intonations. Designed for deployment on AWS and GCP.

Flow Forecast *PyTorch* link
Contributed to an open-source deep learning library for time series forecasting. Offers SOTA models, interpretability metrics, and cloud integration for time-series forecasting tasks.

Awards

Smart India Hackathon, 2020 Finalist (Team Leader) Govt. of India
Led the team to the finals of Smart India Hackathon, 2020 for solving the problem of toxicity classification and reduction in the Social Media for health awareness. July 2020

FlowDB: A new large scale river flow, flash flood, and precipitation dataset NeurIPS
Linked, a novel hourly river flow and precipitation dataset and a second subset of flash flood events with damage estimates and injury counts. December 2020

Captain, CSE Cricket Team BIT Durg
Led the branch Cricket team for the inter-branch cricket tournament for consecutive 2 eyars.

Winner in BIT Software & Hardware Innovation Expo, BITSHINE 2018 BIT Durg
Secured First position in BITSHINE 2018 for a prototype for story and article-sharing web platform for sexually assaulted women and depressed people. November 2018